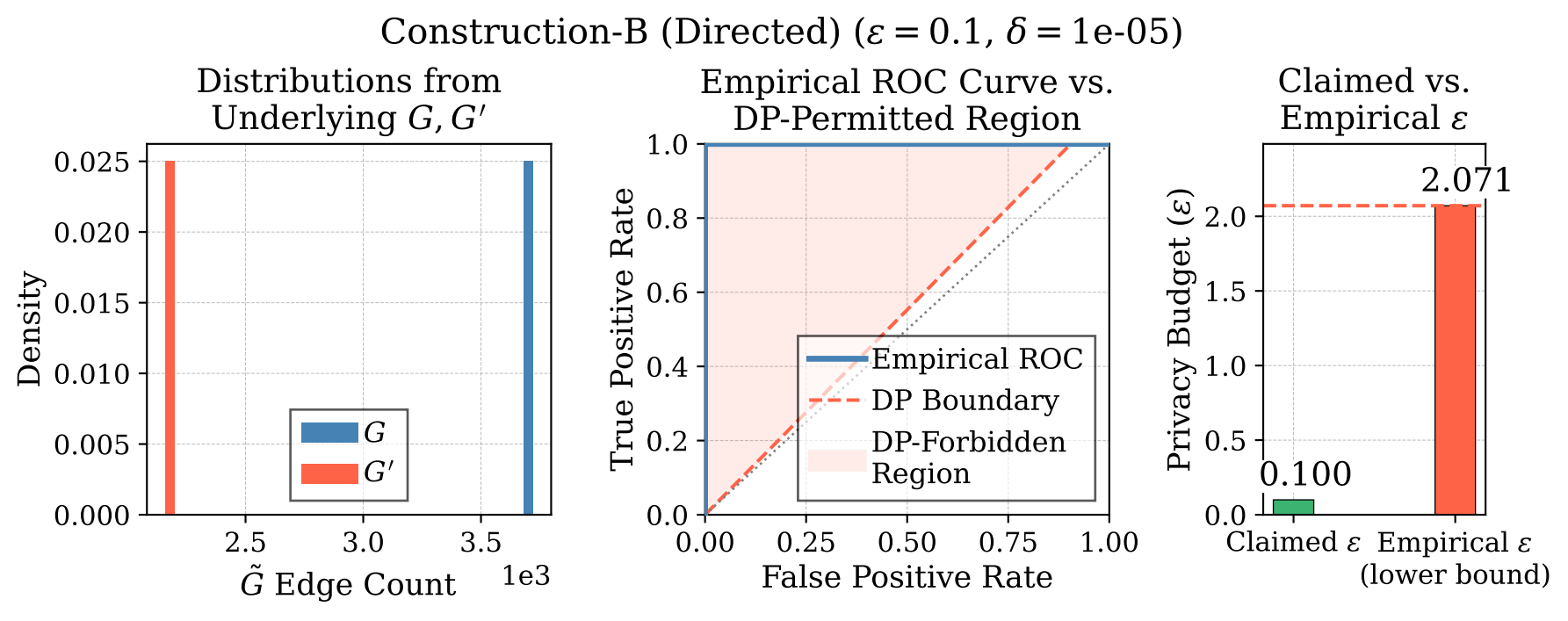
Construction-B (Directed) ($\varepsilon = 0.1$, $\delta = 1\text{e-}05$)
Distributions from Underlying $G, G'$
Density
0.025
0.020
0.015
0.010
0.005
0.000
2.5
3.0
3.5
1e3
$\tilde{G}$ Edge Count
$G$
$G'$
Empirical ROC Curve vs. DP-Permitted Region
True Positive Rate
1.0
0.8
0.6
0.4
0.2
0.0
0.00
0.25
0.50
0.75
1.00
False Positive Rate
Empirical ROC
DP Boundary
DP-Forbidden Region
Claimed vs. Empirical $\varepsilon$
Privacy Budget ($\varepsilon$)
2.071
0.100
2.0
1.5
1.0
0.5
0.0
Claimed $\varepsilon$
Empirical $\varepsilon$ (lower bound)